**Lab 1: Complete Lab-Focused Teaching Package**

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# สมาชิกและหน้าที่

เพชรภิญโญ ธนศิรินรากร 673380073-7 PART 1 DEVICE CONFIGURATION, Bonus Part
ก้องภพ โชควิริยะ 673380030-5 PART 2  CONNECTIVITY VERIFICATION
ณพวิทย์ วงษ์ประเสริฐ 67338266-6 PART 3 ENCAPSULATION ANALYSIS
รัฐภูมิ เกิดพระจีน 673380057-5 Part 4 TROUBLESHOOTING CHALLENGE
อชิรวัช บึงไสย์ 673380298-3 PART 5  ADVANCED OBSERVATION

Source : File

## 🎯 LAB COMPLETION CRITERIA

- ☑ All devices respond to ping from at least one other device
- ☑ ARP tables show correct IP-MAC mappings
- ☑ Switch MAC table shows both PCs' MAC addresses
- ☑ Simulation clearly shows encapsulation process
- ☑ All troubleshooting scenarios resolved
- ☑ Worksheets completed with accurate data

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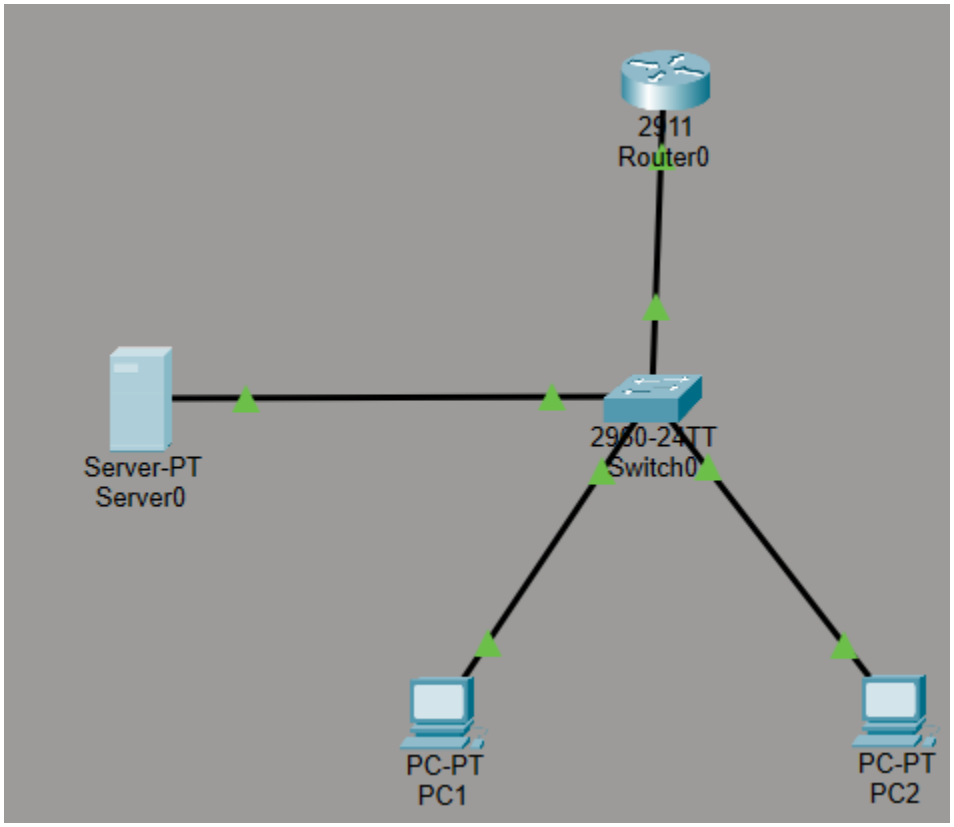
# 📁 PACKET TRACER FILE REQUIREMENTS

**File Name:** `ITN_Module3_ProtocolLab.pkt` **Required Components:**

- 1× Router 2911 (R1)
- 1× Switch 2960 (S1)
- 2× PCs (PC1, PC2)
- Copper straight-through cables
- All configurations from this lab script

✅**Bonus:** Add a server at 192.168.1.100 for HTTP protocol analysis




```
C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.1.1           0001.63e5.be01        dynamic
  192.168.1.2           0060.700b.4339        dynamic
  192.168.1.11          00d0.d3a7.e322        dynamic
  192.168.1.100         00d0.ffd1.c4d0        dynamic
```

Source :
https://drive.google.com/file/d/1HBI21_A_W0342nTYpnocAjTjbsCTJ8Y3/view?usp=sharing

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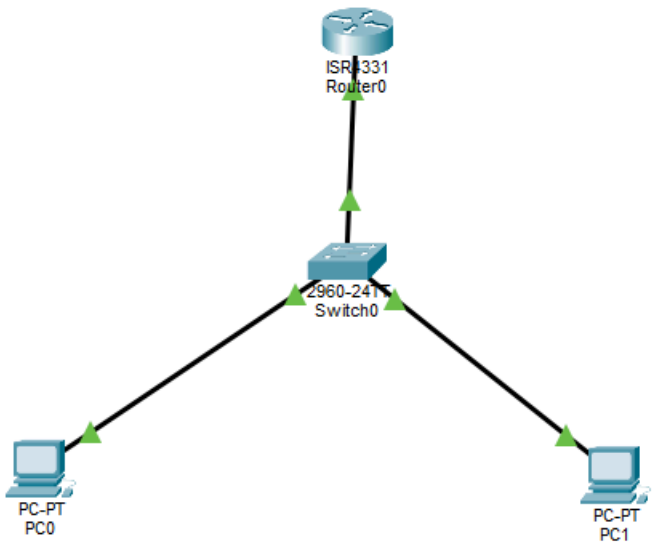
## Part 1: Configure basic device addressing (R1, S1, PC1, PC2)




## ADDRESSING TABLE

| Device | Interface | IP Address | Subnet Mask | Default Gateway | MAC (Dynamic) |
|---|---|---|---|---|---|
| R1 | G0/0/0 | 192.168.1.1 | 255.255.255.0 | N/A | 00D0.BC6A.3501 |
| S1 | VLAN 1 | 192.168.1.2 | 255.255.255.0 | 192.168.1.1 | 00D0.BA1B.D8EC |
| PC1 | NIC | 192.168.1.10 | 255.255.255.0 | 192.168.1.1 | 0060.2F28.0395 |
| PC2 | NIC | 192.168.1.11 | 255.255.255.0 | 192.168.1.1 | 000D.BDC4.3914 |

**Network:** 192.168.1.0/24
**Usable Host Range:** 192.168.1.1 - 192.168.1.254
**Broadcast:** 192.168.1.255

## PART 2: CONNECTIVITY VERIFICATION

### PC1

```
Pinging 192.168.1.11 with 32 bytes of data:

Reply from 192.168.1.11: bytes=32 time=4ms TTL=128
Reply from 192.168.1.11: bytes=32 time=4ms TTL=128
Reply from 192.168.1.11: bytes=32 time=4ms TTL=128
Reply from 192.168.1.11: bytes=32 time=4ms TTL=128

Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 4ms, Maximum = 4ms, Average = 4ms

C:\>
```

```
C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.1.1           00d0.bc6a.3501        dynamic
  192.168.1.2           00d0.ba1b.d8ec        dynamic
  192.168.1.11          000d.bdc4.3914        dynamic
```

### PC2

```
C:\>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.1.1           00d0.bc6a.3501        dynamic
  192.168.1.2           00d0.ba1b.d8ec        dynamic
  192.168.1.10          0060.2f28.0395        dynamic
```

## R1

```
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   192.168.1.1     YES NVRAM  up                    up
GigabitEthernet0/0/1   unassigned      YES NVRAM  administratively down down
GigabitEthernet0/0/2   unassigned      YES NVRAM  administratively down down
Vlan1                  unassigned      YES NVRAM  administratively down down
R1#show arp
Protocol  Address          Age (min)  Hardware Addr   Type   Interface
Internet  192.168.1.1            -   00D0.BC6A.3501  ARPA   GigabitEthernet0/0/0
Internet  192.168.1.10           5   0060.2F28.0395  ARPA   GigabitEthernet0/0/0
Internet  192.168.1.11           2   000D.BDC4.3914  ARPA   GigabitEthernet0/0/0
```

## S1

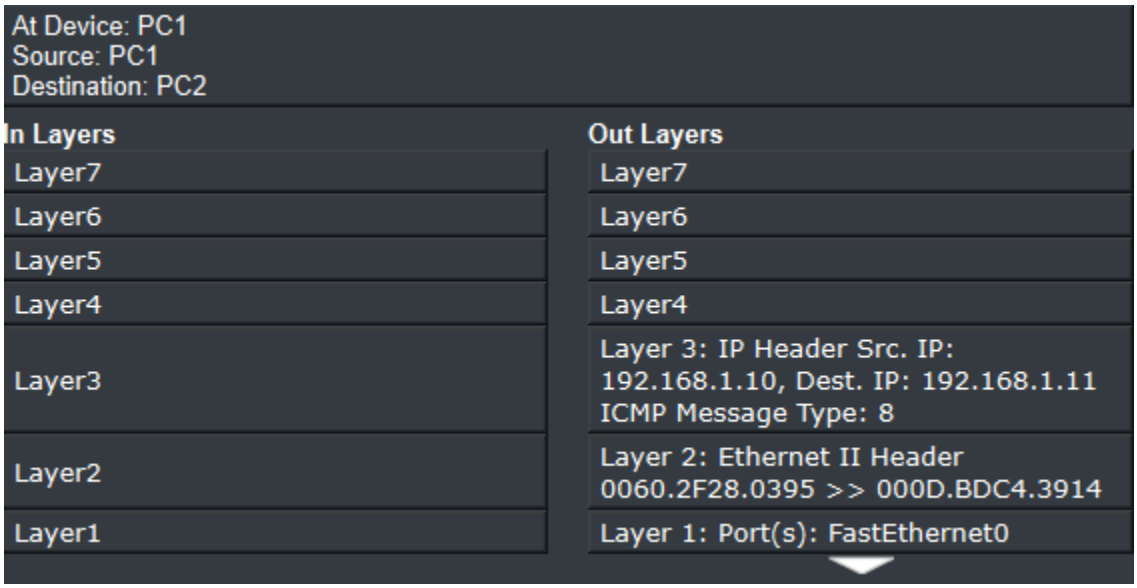
```
S1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
S1#
```

```
Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0060.2f28.0395    DYNAMIC     Fa0/2
   1    00d0.bc6a.3501    DYNAMIC     Fa0/1
S1#
```

## Part 3: Analyze protocol headers using simulation

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| Step | Protocol | OSI Layer | TCP/IP Layer | Key Field Observed |
|---|---|---|---|---|
| 1 | ARP Request | Layer 2 | Network Access | Target IP: 192.168.1.11 |
| 2 | ARP Reply | Layer 2 | Network Access | 000D.BDC4.3914 |
| 3 | ICMP Echo | Layer 3 | Internet | Type: 8, Code: 0 |
| 4 | Ethernet | Layer 2 | Network Access | Src MAC : 0060.2F28.0395 👉<br>Dest MAC : 000D.BDC4.3914 |
| 5 | ICMP Reply | Layer 3 | Internet | Type: 0, Code: 0 |

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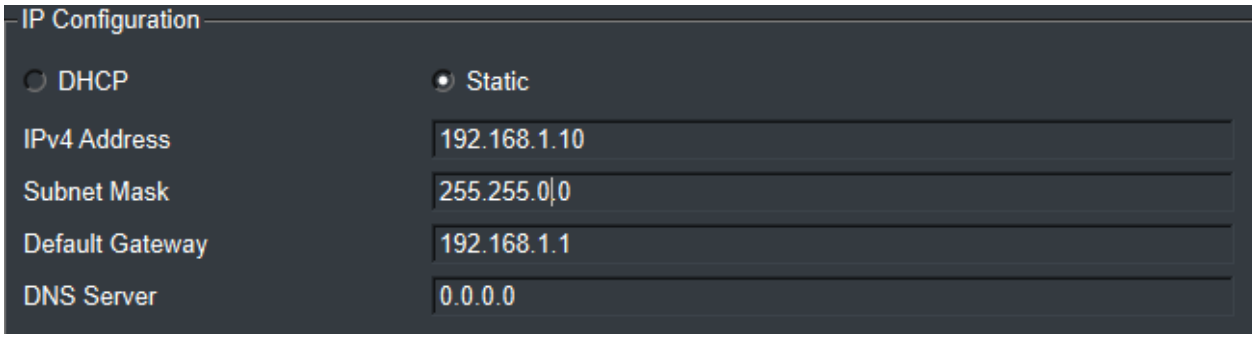
## Part 4: Troubleshoot layered connectivity issues

### Scenario A: Incorrect Subnet Mask

1) On PC1, change to wrong subnet mask:




```
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::260:2FFF:FE28:395
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.1.10
   Subnet Mask.....................: 255.255.0.0
   Default Gateway.................: ::
                                     192.168.1.1

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0
```

2) Ping to PC2

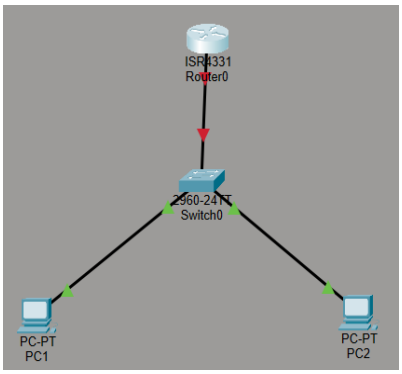
```
C:\>ping 192.168.1.11

Pinging 192.168.1.11 with 32 bytes of data:

Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

3) arp -a

```
C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.1.11          000d.bdc4.3914        dynamic
```

## Scenario B: Router Interface Shutdown

1) ปิดพอร์ตที่ router g0/0/0




```
R1>enable
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   192.168.1.1     YES NVRAM  up                    up
GigabitEthernet0/0/1   unassigned      YES NVRAM  administratively down down
GigabitEthernet0/0/2   unassigned      YES NVRAM  administratively down down
Vlan1                  unassigned      YES NVRAM  administratively down down
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface g0/0/0
R1(config-if)#shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to down
```

2) Test ping from PC1

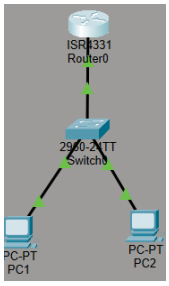
```
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

3) examine ARP table

```
C:\>arp -a
No ARP Entries Found
```

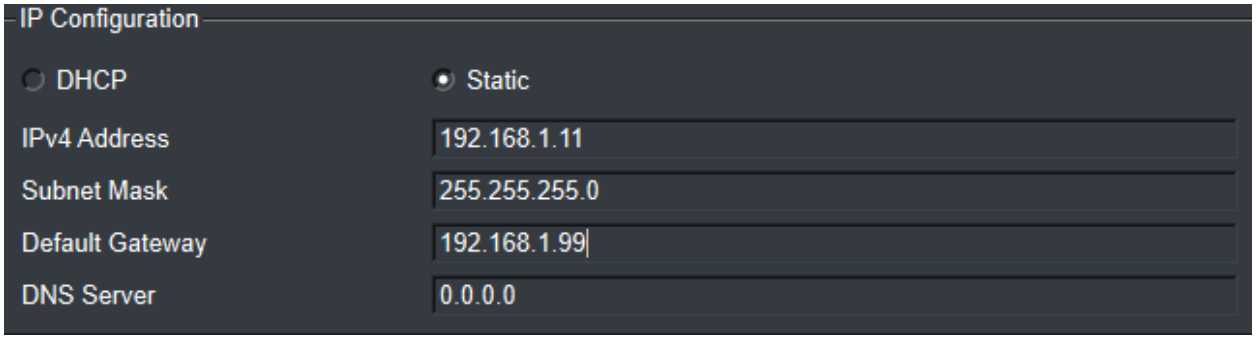
4) Fix: open router

```
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up
```



## Scenario C: Wrong Default Gateway

### 1) Change Default Gateway




```
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::20D:BDFF:FEC4:3914
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.1.11
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     192.168.1.99

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0
```

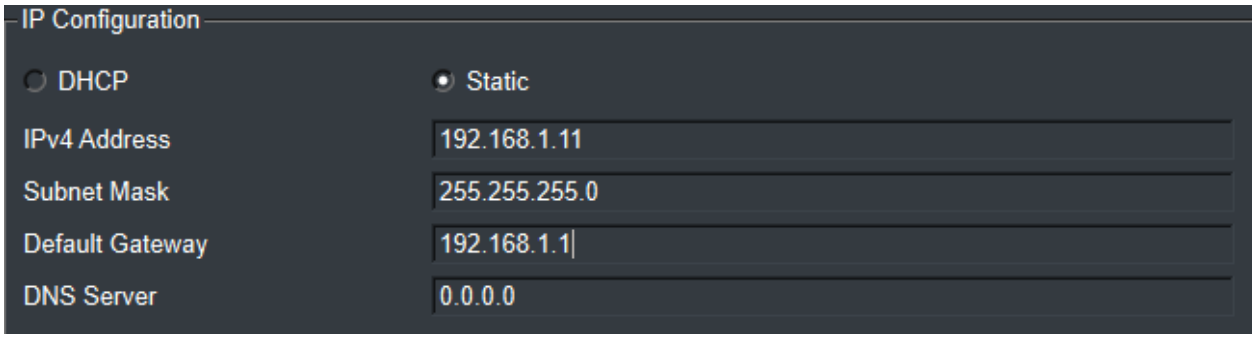
### 2) Ping from PC2 to R1

```
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

### 3) route print

```
R1#enable
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0/0
```

### 4) Fix : reset Default Gateway



## PART 5: ADVANCED OBSERVATION (15 minutes)

### Step 9: Protocol Stack Verification

```
C:\>ping 192.168.1.11

Pinging 192.168.1.11 with 32 bytes of data:

Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>arp -d
C:\>ping 192.168.1.11

Pinging 192.168.1.11 with 32 bytes of data:

Reply from 192.168.1.11: bytes=32 time=475ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 475ms, Average = 118ms

C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.1.11          000d.bdc4.3914        dynamic

C:\>
```

ทำการปิง 192.168.1.11
จากนั้นเคลียร์ arp และปิงอีกรอบเพื่อดูผล

```
S1>enable
S1#clear mac address-table dynamic
S1#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

S1#
```

เคลียร์ mac address-table แล้วแสดดงผลออกมา แล้วไม่มีข้อมูล

แต่หลังจากปิง pc1 to pc2 พบว่ะ

```
S1#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    000d.bdc4.3914    DYNAMIC     Fa0/3
   1    0060.2f28.0395    DYNAMIC     Fa0/2
   1    00d0.bc6a.3501    DYNAMIC     Fa0/1
S1#
```

```
None
! From PC1, generate extended ping with different protocols:
PC1> ping 192.168.1.11 -n 5 -l 1500  ! Test MTU/fragmentation
PC1> arp -d *  ! Clear ARP cache, then ping to observe ARP
process

! On S1, observe MAC table changes:
S1# clear mac address-table dynamic
S1# show mac address-table  ! Observe empty table
! Generate traffic, then check again
```

### Step 10: Documentation

Create a **protocol interaction diagram** showing:

1. Application data generation (simulated)
2. TCP segment creation (if applicable)
3. IP packet encapsulation
4. Ethernet framing
5. Physical transmission

[PC1]

1. Data: "Echo Request"
2. Transport: ICMP
3. Network: Add IP (192.168.1.10)
4. Data Link: Add MAC (0060.2F28.0395)
5. Physical: ส่งสัญญาณออกทางพอร์ต FastEthernet0

[Switch1]

รับ Bits → ตรวจสอบ MAC ใน Frame → ส่งต่อ Bits ออกทางพอร์ตที่เชื่อมกับ PC2

[PC2]

รับ Bits → แกะ Frame (Check MAC) → แกะ Packet (Check IP) → อ่าน Data

## ผลการทดลองและการวิเคราะห์ (Observation & Analysis)

1. กระบวนการ Encapsulation ที่ PC1

Layer 7–5 (Application / Presentation / Session)

- ผู้ใช้สั่งคำสั่ง Ping
- ระบบเริ่มต้นกระบวนการ ICMP

Layer 4 (Transport)

- ICMP ไม่ใช้ Transport Layer (TCP/UDP)

Layer 3 (Network – IP Header)

- Source IP: 192.168.1.10
- Destination IP: 192.168.1.11
- ICMP Message Type: 8 (Echo Request)

PC1 ตรวจสอบพบว่าปลายทางอยู่ในเครือข่ายเดียวกัน จึงไม่ส่งผ่าน Default Gateway

2. กระบวนการ ARP (Address Resolution Protocol)

- PC1 ส่ง ARP Request แบบ Broadcast เพื่อค้นหา MAC Address ของ PC2
- PC2 ตอบกลับด้วย ARP Reply พร้อม MAC Address ของตนเอง
- PC1 บันทึกข้อมูลลงใน ARP Table

3. Layer 2 Encapsulation (Ethernet Frame)

- Source MAC: MAC Address ของ PC1
- Destination MAC: MAC Address ของ PC2
- Frame ถูกสร้างและส่งไปยัง Switch

4. การทำงานของ Switch (Switch Forwarding)

- Switch รับเฟรมที่พอร์ต FastEthernet
- ตรวจสอบ Destination MAC Address
- ค้นหา MAC Address Table
- ส่งต่อเฟรมไปยังพอร์ตที่เชื่อมต่อกับ PC2

Switch ทำงานเฉพาะ Layer 2 ไม่มีการแก้ไข IP Address

5. กระบวนการ Decapsulation ที่ PC2

Layer 2

- ตรวจสอบ MAC Address ตรงกับของตนเอง
- ลอก Ethernet Header ออก

Layer 3

- ตรวจสอบ Destination IP
- พบว่าเป็น IP ของตนเอง
- ส่ง ICMP Echo Request ให้ ICMP Process

6. ICMP Echo Reply จาก PC2

- PC2 สร้าง ICMP Message Type 0 (Echo Reply)
- Source IP: 192.168.1.11
- Destination IP: 192.168.1.10
- Encapsulation ใหม่ทุก Layer

7. การรับ Echo Reply ที่ PC1

- Switch ส่งต่อเฟรมกลับไปยัง PC1
- PC1 ทำ Decapsulation
- รับ ICMP Echo Reply สำเร็จ

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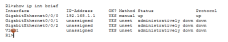
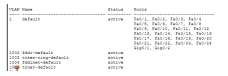
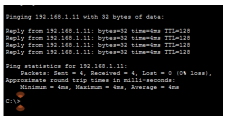
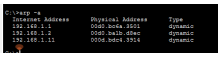
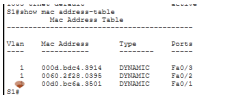
# 📝 LAB DELIVERABLES

## Worksheet 1: Configuration Verification

| Checkpoint | Command Used | Expected Result | Actual Result | Pass/Fail |
|---|---|---|---|---|
| R1 Interface Status | `show ip int brief` | G0/0 UP/UP |  | PASS |
| S1 VLAN Config | `show vlan brief` | VLAN 1 Active |  | PASS |
| PC1 Connectivity | `ping 192.168.1.11` | Success (4 replies) |  | PASS |
| ARP Resolution | `arp -a` on PC1 | 2+ entries |  | PASS |
| MAC Learning | `show mac address-table` on S1 | 2 dynamic entries |  | PASS |

## Worksheet 2: Protocol Analysis

**Capture this data during simulation:**

1. **ARP Request Frame:**

    - Source MAC: 0060.2F28.0395
    - Destination MAC: 000D.BDC4.3914
    - ARP Opcode: :0x0010

2. **ICMP Echo Packet:**

    - Time to Live (TTL): 128
    - Protocol Field: 1 (Value = 1)

3. **Encapsulation Chain:** Data → ICMP Header → IP Header (192.168.1.10/11) → Ethernet Header (MAC Addresses) → Bits

## Worksheet 3: Troubleshooting Journal

| Problem Injected | Symptoms Observed | Diagnostic Commands Used | Root Cause | Solution |
|---|---|---|---|---|
| Wrong Subnet Mask | PC1 ไม่สามารถ ping ไปยัง PC2 | `ping`, `arp -a` | ตั้งค่า Subnet Mask ไม่ตรงกัน | เปลี่ยน subnet ให้เหมือนกัน เป็น 255.255.255.0 |
| Interface Shutdown | Request timeout | `show ip int brief` | Interface g0/0 ที่ R1 ถูก shutdown | ใช้งานคำสั่ง no shutdown ตรงR1 |
| Incorrect Gateway | PC2 ไม่สามารถ ping ไปยัง R1 | `ipconfig`, `tracert` | Default Gateway ถูกตั้งค่าไม่ถูกต้อง | เปลี่ยน Default Gateway ให้เหมือนกัน เป็น 192.168.1.1 |

---

# 🔍 VERIFICATION COMMANDS CHECKLIST

## Router (R1):

☑ `show running-config`

```
R1#show running-config
Building configuration...

Current configuration : 701 bytes
!
version 15.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname R1
```

```
interface GigabitEthernet0/0/0
 description LINK_TO_SWITCH_S1
 ip address 192.168.1.1 255.255.255.0
 duplex auto
 speed auto
!
interface GigabitEthernet0/0/1
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface GigabitEthernet0/0/2
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
```

☑ `show ip interface brief`

```
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   192.168.1.1     YES manual up                    up
GigabitEthernet0/0/1   unassigned      YES unset  administratively down down
GigabitEthernet0/0/2   unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
```

☑ show arp

```
R1#show arp
Protocol  Address          Age (min)  Hardware Addr   Type   Interface
Internet  192.168.1.1             -   00D0.BC6A.3501  ARPA   GigabitEthernet0/0/0
```

☑ show ip route

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0/0
```

## Switch (S1):

☐ show running-config

```
interface FastEthernet0/1
 switchport mode access
!
interface FastEthernet0/2
 switchport mode access
!
interface Vlan1
 ip address 192.168.1.2 255.255.255.0
!
ip default-gateway 192.168.1.1
```

☑ show vlan brief

```
S1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

☑ show mac address-table

```
S1#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    000d.bdc4.3914    DYNAMIC     Fa0/3
   1    0060.2f28.0395    DYNAMIC     Fa0/2
   1    00d0.bc6a.3501    DYNAMIC     Fa0/1
```

- ☑ `show interfaces status`

```
S1#show interface status
Port      Name               Status       Vlan       Duplex  Speed Type
Fa0/1                        connected    1          a-full  a-100 10/100BaseTX
Fa0/2                        connected    1          a-full  a-100 10/100BaseTX
Fa0/3                        connected    1          a-full  a-100 10/100BaseTX
```

## End Devices:

- ☑ `ipconfig /all` (or equivalent)

PC1

```
C:\>ipconfig /all

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Physical Address................: 0060.2F28.0395
   Link-local IPv6 Address.........: FE80::260:2FFF:FE28:395
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.1.10
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     192.168.1.1
   DHCP Servers....................: 0.0.0.0
   DHCPv6 IAID.....................:
   DHCPv6 Client DUID..............: 00-01-00-01-A1-D3-8C-4E-00-60-2F-28-03-95
   DNS Servers.....................: ::
                                     0.0.0.0

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Physical Address................: 000D.BDD7.A201
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0
   DHCP Servers....................: 0.0.0.0
   DHCPv6 IAID.....................:
   DHCPv6 Client DUID..............: 00-01-00-01-A1-D3-8C-4E-00-60-2F-28-03-95
   DNS Servers.....................: ::
                                     0.0.0.0
```

PC2

```
C:\>ipconfig /all

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Physical Address.................: 000D.BDC4.3914
   Link-local IPv6 Address.........: FE80::20D:BDFF:FEC4:3914
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.1.11
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     192.168.1.1
   DHCP Servers....................: 0.0.0.0
   DHCPv6 IAID.....................:
   DHCPv6 Client DUID..............: 00-01-00-01-26-D4-DC-75-00-0D-BD-C4-39-14
   DNS Servers.....................: ::
                                     0.0.0.0

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Physical Address.................: 0050.0F8B.9C36
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0
   DHCP Servers....................: 0.0.0.0
   DHCPv6 IAID.....................:
   DHCPv6 Client DUID..............: 00-01-00-01-26-D4-DC-75-00-0D-BD-C4-39-14
   DNS Servers.....................: ::
                                     0.0.0.0
```

☑ `arp -a`

```
C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.1.1           00d0.bc6a.3501        dynamic
  192.168.1.2           00d0.ba1b.d8ec        dynamic
  192.168.1.11          000d.bdc4.3914        dynamic
```

```
C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.1.1           00d0.bc6a.3501        dynamic
  192.168.1.2           00d0.ba1b.d8ec        dynamic
  192.168.1.10          0060.2f28.0395        dynamic
```

☑ `ping` to all devices

```
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=255
Reply from 192.168.1.2: bytes=32 time<1ms TTL=255
Reply from 192.168.1.2: bytes=32 time<1ms TTL=255
Reply from 192.168.1.2: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.11

Pinging 192.168.1.11 with 32 bytes of data:

Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time=1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

```
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=255
Reply from 192.168.1.2: bytes=32 time<1ms TTL=255
Reply from 192.168.1.2: bytes=32 time<1ms TTL=255
Reply from 192.168.1.2: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time=8ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 8ms, Average = 2ms
```

☑ `tracert` (if multi-hop)

---

## 🎓 ASSESSMENT QUESTIONS

1. When PC1 pings PC2, which protocol resolves the MAC address before ICMP operates?
    - ARP เพราะในระดับ Layer 2 การจะส่งข้อมูลได้ต้องใช้ MAC Address แต่ PC1 รู้เพียง IP ของ PC2 มันต้องส่ง ARP Request ก่อนที่ ICMP จะเริ่มทำงานได้
2. What would happen if you changed the subnet mask on PC1 to 255.255.255.128?
    - Pinh หากันได้
3. Why does the switch not need an IP address for basic frame forwarding?
    - เพราะ Switch (Layer 2) ทำงานโดยใช้ MAC Address Table
4. Which OSI layers are involved when R1 forwards a packet between interfaces?
    - Layer 1, Layer 2 และ Layer 3
5. What is the purpose of the TTL field in the IP header you observed?
    - ทุกครั้งที่แพ็กเกจวิ่งผ่าน Router หนึ่งตัว ค่า TTL จะถูกลดลงทีละ 1 หากค่า TTL ลดลงจนเหลือ 0 แพ็กเกจนั้นจะถูกทำลายทิ้งทันที เพื่อไม่ให้เครือข่ายบวมหรือล่มจากการที่มีข้อมูลวิ่งค้างอยู่ไม่สิ้นสุด

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